

Name: **SANTONU SARKAR**

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DOB: 5/5/1966 Nationality: Indian

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EXECUTIVE SUMMARY

- Over 30 years of experience with a unique blend of in-depth academic research, goal-oriented industrial research, hands-on complex software development (C++, Java, Python, C#), and architecture consulting. Demonstrated leadership in client management and business development
- Strong research expertise in the cloud and edge based, service-oriented system design as well as science of classical software development. Worked extensively in the areas of industrial automation, dependability of SAAS and service-oriented systems, software architecture, modularity, automated software engineering. Worked extensively on developing machine learned systems.
- Built research groups in i) software engineering ii) dependability of cloud based systems and services, in collaboration with Purdue University, the University of Illinois at Urbana Champaign and University of Naples. Published a book chapter on the dependability of IOT platforms.
- Published 70+ papers in peer-reviewed journals, national and international conferences, 17 US patents in software engineering and systems
- Worked on anomaly detection of service-oriented, cloud deployed systems in collaboration with the University of Illinois at Urbana Champaign and University of Naples. Developed novel machine learning methods to identify anomalies from various types of data.
- Have 8+ years of experience in knowledge creation and dissemination using academic pedagogy. Delivered 20+ national and international invited talks, tutorials and keynotes at various educational institutions and conferences.

EDUCATION

1992–1995 Ph.D.	Computer Science	Indian Institute of Technology, Kharagpur	
	· Title: An Object Oriented Approach to Digital Circuit Synthesis		
	· http://www.idr.iitkgp.ac.in/xmlui/handle/123456789/8679		
1990–1992 M.Tech	Computer Science	Indian Institute of Technology, Kharagpur	9.43/10
1984–1988 Bachelor in Engineering	Computer Science	Indian Institute of Engineering Science & Technology, Shibpur	82% 1st rank in class

ACADEMIC EXPERIENCE

01/2023 – 06/2025	Professor and Head, Dept of Computer Science and Information Systems, Birla Institute of Technology and Science Pilani - K.K.Birla Goa Campus, India
	· <i>Courses Taught:</i> Distributed Computing, Cloud Computing, Compiler Design
	· <i>Administration:</i> Department chair from 2023-2025. Responsible for managing the department, faculty recruitment, performance management, academic and research vision of the department and student affairs. Streamlined academic processes through automation, such as course offerings, teaching assistantship, document approval and student evaluation. Introduced a new masters program in Quantum Information Processing. Nucleus member of the Institutional Knowledge Analysis Cell.
	· <i>Research:</i> Working on developing a framework for real-time condition monitoring and optimization for diverse domains. Developed a digital twin framework for crop yield prediction and disease spread simulation using reinforcement learning. Working on developing machine learning based anomaly detection- targeted for real time monitoring.
07/2014–11/2019, 06/2025–ongoing	Professor, Dept of Computer Science and Information Systems, Birla Institute of Technology and Science Pilani - K.K.Birla Goa Campus, India
	· <i>Courses Taught:</i> Software Architecture, Software Engineering, Cloud Computing and cyber-physical systems, Compiler Design, Principles of Programming Languages, Programming Language and Compiler Construction
	· <i>Administration:</i> Head of the department level selection committee for faculty recruitment. Nucleus member of graduate study committee.
	· <i>Research:</i> Worked on dependability of SAAS applications using data-driven approaches. Worked on anomaly detection of service-oriented, cloud deployed systems in collaboration with the University of Illinois at Urbana Champaign and University of Naples. Developed novel machine learning methods to identify anomalies from various types of data.
	· Worked on SLA aware meta-cloud platform. Worked on performance and power prediction of GPU applications.
	· Working on developing smart and sustainable mobility solutions- such as altruism based ride sharing, real-time monitoring of driving patterns.
	· Working on developing a framework for sustainable HPC such as real-time temperature management, energy-efficiency and performance prediction.

	- Working on neurosymbolic computing - developing language model and formal verification based approach for software evolution. - Working on developing a digital twin framework for urban water distribution networks.
09/2018–09/2019	Visiting Professor, IIIT - Bangalore
01/1995–02/1996	<i>Courses Taught:</i> Advanced Compiler (Optimization and Parallelization)
	Lecturer, Computer Science & Tech., Indian Institute of Engineering Science & Technology - Shibpur India
	<i>Courses Taught:</i> Digital Circuits, Data Structure, VLSI Design
1992–1994	Junior Scientific Officer and Research Assistant, Dept of CSE, IIT Kharagpur
	<i>Teaching Assistant:</i> Introduction to Computing, Data structure, OS Design lab, Compiler design lab, Database lab <i>Project:</i> Key designer and developer of a fuzzy rule based expert system for Defense Electronics Application Laboratory (DEAL) India. <i>Achievements:</i> Resulted in more research projects sponsored by DEAL. The expert system was general enough to be applied in other domain such as Medical Diagnosis.
1989–1990	Programming Assistant, Dept of CSE, IIT Kharagpur
	Involved in implementation of a Hardware Description language (KIDLAN) compiler for hardware circuit description and synthesis for Department of Electronics India. Developed in HP-UX, C, curses.h package, lex and yacc.

SPONSORED PROJECTS

2014-2018	Addressing Challenges in Application Development for Heterogeneous Computing Platform for Speedup and Power Efficiency, <i>SERB, Govt of India</i>	INR 37.55L	Completed	PI
2015-2018	A Cost-effective Approach of Making Software Dependable, <i>Accenture Technonology Labs</i>	USD 25K	Completed	PI
2016-2018	Applications of High Performance Computing to Software Engineering and Data Analytics, <i>BITS Pilani</i>	INR 14L	Completed	PI
2024-2025	Ensuring Food Security through AI and Satellite Imaging Technology assisted Robust Agriculture at Scale, <i>CPMU & IIT Jammu</i>	INR 1.98CR	Completed	Co-PI
2023-2025	Developing Advanced Membrane System Design Software, <i>ION Exchange</i>	INR 7.2L	Ongoing	Co-PI
2023-2025	Edge Analytics Solution For Autonomous Condition Monitoring Of Industrial Systems, <i>BITS Pilani New Faculty Seed Grant</i>	INR 18L	Completed	PI
2025-2030	Physics-Informed Digital Twin for Urban Water Distribution Networks, <i>Wipro Research</i>	INR 50L	Ongoing	PI

INDUSTRY EXPERIENCE

12/2019–01/2023	Senior Principal Scientist, ABB Corporate Research - India - Developed a validator for modular, industrial process control system specification (C#, Rule Engine, ECORE metamodel, AutomationML) - Developed a prototype for validating safety properties of modular industrial components - Led an India-Germany collaborative research project on distributed Edge platform for industrial automation (Azure IoT, C#, Python, MQTT, ML techniques). Worked on developing a hierarchical edge platform for industrial automation, for analytics application deployment. - Developed a federated learning prototype on Azure based Hierarchical Edge platform. - Global leadership role: i) Technology strategy and vision (Software and Control) ii) University relations
01/2018–12/2018	Principal Research Consultant (Sabbatical from BITS Pilani), ABB Corporate Research - India - Led a team of size three to migrate a monolithic industrial application to a containerized environment and analyzed the challenges thereof. Responsible for managing multi-country business stakeholders. - Windows 10, C#, WCF, dotNET, Docker and AMQP.
05/2010–07/2014	Principal Research Scientist, Infosys Labs - India Role: Head, Next Generation Computing Research group. - Responsible for building and leading cloud and HPC research group. Managed a research team of size 7-10. - Responsible for building defining research vision, creating research solution, and foster research collaboration with internal and external partners. <i>Research:</i> i) Investigated research challenges in building Dependable Systems in Cloud and Virtualized Platform ii) Developed framework for Porting Applications for Multi-core, GPGPU Computing <i>Objective:</i> To address dependability and performance challenges in cloud based, distributed (like Hadoop) and GPGPU applications

- Built a corporate sponsored “dependability center of excellence”, to develop data-driven, modeling and prediction tools to help practitioners assess and analyze a “Dependability issues” of applications. Created a new consulting service model on Dependability assessment for the organization
 - Deployed solutions to various SaaS platforms, and actively involved in consulting, pre-sales of the new service-line with the client facing group to promote research outcome.
- 11/2007–
05/2010
- VP and Research Lead, Accenture Technology Labs - India**
Role: Lead, Software Engineering Research
- Responsible for applied research based asset development, manage the India lab research group,
 - Build research network and ecosystem within and outside the firm, manage research hiring and global internship program (such as MIT Accenture relationship program). Filed several patents and published papers in the area of software design.
 - *Research Area:* i) Software Design and ii) Application Maintenance
 - Conceptualized a long term research agenda applicable during software design and maintenance. Initiated a umbrella of projects to develop a tool to perform i) Design recovery using architecture description DSL ii) Impact analysis iii) Design automation and iv) Design quality analysis v) Application Maintenance Optimization program .
- 09/2002–
11/2007
- Principal Architect, Infosys Labs - India**
Role: Senior researcher in the area of software engineering
- Managed a research team of size 6. Responsible for building research collaboration.
 - Also involved in several delivery projects as architecture consultant, such as Architecture assessment and recommendation of Banking Product and Architecture assessment and performance analysis (workload modeling, network bandwidth analysis) of a J2EE based financial application in Chicago.
 - Provided proposal & Presales support for more than 5 large multi-year projects.
 - *Research Area:* Large System Comprehension & Modularization
 - Invented a set of new metrics to measure modularity of software. Build toolset to measure and diagnose the modularity problems exist in the software. Worked on measuring architectural layering violation and automatic discovery of architectural layers. Worked on identifying functional/business concepts from source code using machine learning technique like LDA.
 - *Research Area:* Software Architecture
 - Conceptualized, led and deployed a research project on Application Architecture Methodology – that involves definition and deployment of an architecture process, definition and implementation of an architecture modeling language. The modeling language is influenced by the standards like Component-connector formalism, IEEE 1471, MDA. Involved in the capacity of a mentor in Enterprise Architecture Solution offering.
- 03/2002–
09/2002
- Senior Tech Lead, Hughes Software Systems - Bangalore India and Germantown MD USA**
Role: Worked as the architect in building DirecwayTM wireless internet access service for Hughes Network Systems (Germantown, USA).
- The service is built in Java (JDK 1.3) and uses Weblogic 6.1 server & Oracle 8.1 database, RADIUS server for network authentication and USG devices.
 - Involved in requirement analysis, architecture definition, a part of user interface design, low level design and prototyping. Also acted as on-site coordinator for the Hughes off-shore team.
 - *Achievements:* More downstream projects at Hughes Bangalore. Successful launching of the final product at various commercial outlets
- 01/2002–
03/2002
- Programmer Analyst, TCG Software Services - Edison NJ, USA**
Role: Business Analyst
- Client Interaction, requirement collection, high level design of a genetic banking project
- 06/2001–
12/2001
- Senior Software Developer, Encoda Systems - Boston MA, USA**
Role: Technical lead in building an ad-commerce product (MOS) for Encoda Systems (Boston USA) for the media industry.
- Built using JDK 1.3, Weblogic 5.1 server, XML, Oracle 8.1 database and SONIC MQ.
 - Involved in product feature analysis, design & coding of the current version of the product.
 - *Achievements:* Development, build and release process streamlined. Initiated re-architecting the product for the future version
- 08/1999–
05/2001
- Technical Architect, EFORCE Inc. - Boston MA, USA**
Role: Project consulting- Discovery and scope definition. Also involved in product architecture
- Requirement gathering and design of the ClaimPlace Claim Repository system for Hartford Insurance USA.
 - Involved in project planning, architecture design, detail design and coding (MVC framework, EJBs, weekly build process streamlining) of the system. For a 401K portal project, developed the back-end framework development to integrate external entities like Banks, NSCC.
 - Also took part in several architecture consulting projects.

- Was a key member of the Databroker server (CORBA based high performance, multi-threaded JDBC1.0 type-3 driver- the flagship product of I-Kinetics/eForce) 7.0 design team. Designed and developed pooling, life-cycle model (resembles EJB object's life cycle) and event handling framework. Main contributor of Databroker 7.x framework design. Developed in NT, VC++5.0, Roguewave tools (dbtools and tools), Orbix 2.3 orb, Oracle 8.0.

- *Achievements:* Quick prototyping and Early product evaluation strategy helped to make significant architectural decision at the early stage of the project. This resulted in minimizing the productivity loss which was appreciated by the client.

07/1997–
07/1999 **Associate Consultant, Wipro Ltd - Bangalore India and Tokyo Japan**

Role: Acted as Solution Architect for the Trade Confirmation System development for Goldman Sachs (Tokyo, Japan) and subsequently an informal account manager role.

- Involved in Requirement understanding, Functional design, project planning, system architecture design and implementation of the Trade confirmation system.

- Designed and developed the multi-threaded confirmation server.

- Developed in Sun Solaris, C++ using Orbix2.3, Actuate Reporting Server, Sybase 11 Database.

- *Achievements:* This project was considered as one of the strategic projects for business growth in Japan. This project helped in getting three more projects for the organization and the account grew rapidly.

02/1996–
06/1997 **Project Leader, Lason Inc (Formerly Vetri Software) - Madras India, and Helsinki, Finland**

Role: Led a team to develop an Archive and Retrieval System for MERITA Bank at Finland, which served as a repository for information gathered over the past ten years.

- Involved in the design and implementation of the overall system architecture, Object Model and O/R Mapping for the system, primary and secondary archive server, multi-threaded query server.

- Coordinated user manual & reference manual preparation with the documentation team.

- Developed in NT using VC++2.0, SQL Server 6.5, MS Access database and iXOS Jukebox server and Image Gear 6.0.

- *Achievements:* Successful deployment of the project influenced the organization to embark on productization of the application.

08/1988–
02/1989 **Systems Executive, Uptron India Ltd - India**

- Responsible for supporting proprietary COBOL software running on Uptron Personal Computer.

OPEN-SOURCE PROJECTS

Program Analysis and Verification (Antarbhukti) – <https://github.com/soumyadipcsis/Antarbhukti-LLM> (2024)

- These projects focus on analyzing and verifying the behavior of PLC software systems using Petri net models.

Auto-Tuning of Tenstorrent AI Hardware – <https://github.com/santonus/tenstorrent> (2025)

- identify optimal tuning parameters to execute ML algorithms on Tenstorrent hardware.

Analysis of Bigscholarly Data – <https://github.com/santonus/bigscholarlydata> (2014-2019)

- Collection of projects to crawl, extract topics, and perform predictive analytics on big scholarly data

- Data crawled from ACM/IEEE, AMiner, and Microsoft academic graph

CUDA Microbenchmark – <https://github.com/santonus/benchmark> (2017)

- A set of microbenchmark programs for CUDA

Data analysis for vehicular safety – https://github.com/santonus/iot_safetyanalysis (2018)

- Analysis of publicly available vehicle accident data. A part of my student's masters thesis, jointly supervised from ABB Corp. Research

Dependability Analysis from application and system logs – <https://github.com/santonus/loganalysis> (2017)

- Set of scripts for fault analysis from logs

Predictive Model for Energy Consumption – <https://github.com/santonus/energymodel> (2016-2019)

- Analyses CUDA programs to predict its execution time and energy consumption

Program verification tool – <https://github.com/santonus/codeverification> (2016-2017)

- Python based program verification tool for data race detection in a CUDA program

Library for measuring power and energy consumptions – <https://github.com/santonus/energymeasure> (2016)

- C++ based library called UPAPPI for measuring power and energy consumption of HPC (CUDA, OpenMP) applications

Design Abstraction Library – <https://github.com/santonus/designabstraction> (2016-2019)

- Extends Thrust library to use shared memory, extends library for 2D data, and heterogeneous devices

Equivalence Checking – <https://github.com/santonus/equivchecker> (2015-2016)

- This project analyses two programs to verify their behavioural equivalence.

SERVICES

2025	• Program Co-Chair, IEEE International Conference on Cloud Computing (CLOUD) 2024,2025 • Standardization Committee- IEEE 2975.3 Industrial AI at the Edge - 2023 onwards
2021	• Program Committee, 23rd ACM/IFIP International Conference Middleware 2022 • Program Committee- 36th IEEE/ACM International Conference Automated Software Engineering Workshop • Reviewer Journal "Software Practices and Experiences" • Vice Chair ACM ISoft
2020	• Program Committee, 12th ACM ISEC, 2021 • Program Committee- 25th IEEE Pacific Rim International Symposium on Dependable Computing Main track • Program Committee- 14th Intl Conf. on Software Technologies - Main track
2019	• Program Chair, 12th ACM ISEC, 2019 • Program Committee- 26th IEEE Asia-Pacific Software Engineering Conference Main track • Program Committee- 49th IEEE/IFIP International Conference on Dependable Systems and Networks (DSN2019), Main and Industry Track • Program Committee- 14th Intl Conf. on Software Technologies - Main track
2018	• Organizing Chair, IFIP WG10.4 Dependable Computing and Fault-Tolerance, 73rd Meeting at Goa India • Program Committee- 25th IEEE Asia-Pacific Software Engineering Conference Main track
2017	• Program Chair, 10th ACM ISEC, LNMIIT Jaipur • Associate Editor, Sadhana, Springer Journal (https://www.springer.com/engineering/journal/12046) • Steering committee chair, ISEC • Program Committee- 47th IEEE/IFIP Intl Conf. on Dependable Systems and Network, Industry track; 24th IEEE Asia-Pacific Software Engineering Conference Main track • Org Committee- 2nd Workshop on "Software Engineering Methods for Parallel and High Performance Applications", collocated with 25th ACM Symposium on High-Performance Parallel and Distributed Computing, Washington DC
2016	• General Chair, and PC committee 9th ACM India Software Engineering Conference • Org Committee- Workshop on "Software Engineering Methods for Parallel and High Performance Applications", collocated with 25th ACM Symposium on High-Performance Parallel and Distributed Computing-Tokyo • Program Committee- The 3rd IEEE International Workshop on Reliability and Security Data Analysis (RSDA 2016) is co-located with the 46th Annual IEEE/IFIP International Conference on Dependable Systems and Networks
2015	• General Co-chair, 16th Intl Conference on Distributed Computing and Networking • Program Committee, The 22nd Asia Pacific Software Engineering Conference 2015 • Student Symposium Reviewer, 22nd IEEE Conf. on High Performance Computing (HiPC)
2014	• Program Committee Member • 6th International Workshop on Software Aging and Rejuvenation, collocated with ISSRE • 37th Intl. Conference on Software Engg (ICSE)- NIER Track • Intl. Conference on Program Comprehension • 2nd IEEE International Workshop on Reliability and Security Data Analysis • 6th IEEE International Workshop on Software Aging and Rejuvenation • 25th IEEE International Symposium on Software Reliability Engineering, Industry track
2008–2014	• PhD forum co-chair in Intl Conference on Distributed Computing and Networking • Tutorial and Workshop chair in ACM India Software Engineering Conference • Industry Workshop Organizer in IEEE Intl Symp. On Software Reliability Engg- 2009 • Ad-hoc reviewer in IEEE TSE, IEEE TSC, IEEE TCC, JSS, IEEE SW

ACHIEVEMENTS

Till 2026	Citations: 3078, h-index: 27, i10 index: 56
2026	Best Paper Award, Main track of 21st Intl Conf. on Evaluation of Novel Approaches to Software Engg. (ENASE 2026)
2025	Best Poster Award, 35th IEEE HPDC Conference, 2025
2021	Best Paper Award, Main track of 34th IEEE International Conference on Industrial, Engineering Other Applications of Applied Intelligent Systems (IEA/AIE)
2020	Student Mentor, first prize at SRC@PLDI https://pldi20.sigplan.org/track/pldi-2020-Student-Research-Competition
2018	ACM-India Test of the Time Award for a 2008 publication

2014 onward	Recipient of Outstanding Potential for Excellence in Research and Academics award from BITS Pilani
2007	Key contributor to form global internship program at Infosys, won best mentor award
2006	Recipient, Infosys excellence award for innovation

COLLABORATION

2024	• Collaboration with CDAC Bangalore (HPC, Systems and Quantum Information Processing) • Collaboration with Quidich Innovation Labs (Sports Analytics) • Collaboration with Tenstorrent Technologies (Optimizing Performance on AI-specific hardware)
2017	• Visiting researcher at Univ. of Illinois at Urbana Champaign
2015–2018	• Research collaboration with Siemens corporate research India
2012–2014	• Research Collaboration with University of Illinois at Urbana Champaign • Research Collaboration with Univ of Naples, Italy
2009	• Managed MIT-Accenture relationship program as a part of Accenture global internship program
2005–2007	• Managed Research collaboration with Purdue University

PORTFOLIO

GitHub Portfolio	https://github.com/santonus/
DBLP	https://dblp.uni-trier.de/pers/hd/s/Sarkar:Santonu
Google Scholar	http://scholar.google.co.in/citations?user=8UNP6BIAAAAJ
Linkedin	https://in.linkedin.com/in/santonusarkar

THESIS SUPERVISION

2026 onwards	PhD supervisor, (student's affiliation: BITS Pilani K K Birla Goa Campus, India) • Ongoing Research Area:<i>Adaptive Framework for Context-Aware Behavioral Risk Assessment in Dynamic Driving Environments</i> • Ongoing Research Area:<i>A Physics-informed Digital Twin Framework for Urban Water Distribution Networks</i> • Ongoing Research Area:<i>Neurosymbolic Computing for Software Evolution</i>
2025	Master's Thesis supervisor, (student's affiliation: BITS Pilani K K Birla Goa Campus, India) • Thesis title:<i>Simulation of Areca Nut Plantation Dynamics Using Multi-Agent Digital Twin Framework</i>
2023	PhD supervisor, (student's affiliation: BITS Pilani K K Birla Goa Campus, India) • Thesis title:<i>Prediction of Performance and Power Consumption of GPGPU Applications</i>
2023	PhD co-supervisor, (student's affiliation: BITS Pilani K K Birla Goa Campus, India) • Thesis title:<i>An Investigation of Various Methodologies for Analysis and Prediction of Operational Failures in Industrial Systems</i>
2020	PhD supervisor, (student's affiliation: BITS Pilani K K Birla Goa Campus, India) • Thesis title:<i>Meta Clouds and Meta Services: Constructing an Abstraction Layer Over Multi-clouds</i>
2018	Master's Thesis supervisor, (student's affiliation: BITS Pilani K K Birla Goa Campus, India) • Thesis title:<i>Predicting and Analyzing Risky Behaviour in Traffic Accidents</i>
2016	Master's Thesis supervisor, (student's affiliation: BITS Pilani K K Birla Goa Campus, India) • Thesis title:<i>Refactoring GPGPU Kernels For Optimized Performance & Energy Efficiency</i>
2016	PhD Thesis co-supervisor, (student's affiliation: IIT Bombay-Monash Academy, India) • Thesis title:<i>Improving Energy Efficiency of MapReduce Systems</i>
2014	Master's Thesis supervisor, (student's affiliation: IIIT Bangalore, India) • Thesis title:<i>Mining Flow and State based Invariants from SaaS Application Logs</i>
2014	Master's Thesis supervisor, (student's affiliation: University of Naples, Italy) • Thesis title:<i>A Fault Injection Tool For Java Software Applications</i> • Thesis title:<i>Mining Invariant Relationships for Failure Analysis of Batch Software Systems</i> • Thesis title:<i>On the Use of Text-weighting Schemes to Detect Anomalies from Security Alerts of a Cloud System</i>

PUBLIC APPEARANCES

2020	• Invited Talk: “Simulation and modeling of industrial cyber-physical systems” – Scientific Russia-India Webinar on Cyber-Physical Systems: Society 5.0
2018	• Invited Talk: <i>Modeling Operational Fairness of Hybrid Cloud Brokerage</i> – Purdue University, IN USA • Invited Talk: – ABB Corporate Research, NC USA

- 2017
 - Keynote Talk: *Dependability Issues in Cyber-physical Systems* – IEEE CCEM Pre-Conference Workshop and 2nd Symp on Application of Formal Methods for Safety & Security of Critical Systems, India
 - Invited Talk: *Making Today's Software Resilient* – Ashoka University, India
 - Invited Talk: *Software Development Challenges in Heterogeneous Computing Platforms* – LNMIT, Jaipur
 - Invited Talk: *Building Dependable Software* – Ericsson Research, Bangalore
 - Invited Talk: *Mining Operational logs for Anomaly Detection* – CSL, UIUC USA and SUTD, Singapore
- 2016
 - Invited Talk: *Recent advances in Software Engineering Techniques for Heterogeneous Platform* – SVNIT, Surat
- 2015
 - Invited Talk: *Dependability Assessment of Operational Systems* – ABB Corporate Research India
 - Invited Talk: *Recent advances in Software Engineering Techniques for Heterogeneous Platform* – **Think Parallel**: Workshop on Parallel Computing for Heterogeneous Platform at Siemens India
- 2014
 - Tutorial: *Making applications ready for HPC platform- software engineering challenges and techniques* – ACM Compute Conference, India
 - Invited Talk: *Energy Efficient Distributed Computing* – Indo-Chile workshop on Big Data, BITS Pilani K.K.Birla Goa Campus
 - Tutorial: *Software Fault-tolerance for NextGen Systems: Milieu of models and methods* – 36th Intl. Conference on Software Engineering Conference, Hyderabad India
- 2013
 - Tutorial: *Virtualized Environments- Benefits and Overheads* – 14th Intl Conference on Distributed Computing and Networking TIFR Mumbai
 - Tutorial: *Towards Securing The Smart Grid (In collaboration with CSL and Information Trust Institute, UIUC)* – 26th IEEE International Conference on VLSI Design, Pune India
- 2012
 - Invited Talk: *Software Engineering Challenges in Next Generation Computing Platform: HPC@Infosys* – AMD Labs, Bangalore India
- 2010
 - Invited Talk: *Software Metrics and Measurements* – Siemens Labs Bangalore
- 2009
 - Invited Talk: *Modular Design- Metrics for Software that Ceases to be Modular* – SAP Labs, Sophia Antipolis France
 - Invited Talk: *Role of Modeling in Product Line Architecture* – 2nd ACM India Software Engg. Conference, Pune India
- 2008
 - Invited Talk: *Are software as a service (SAAS) and Cloud Computing the Future?* – 1st ACM India Software Engg. Conference, IIIT Hyderabad India
 - Invited Talk: *Collaboration and Communication Tools for Agile Development* – Accenture Shaping the Future Workshop
- 2007
 - Invited Talk: *Software Engineering Challenges- IT Services Industry Perspective* – 29th Intl. Conference on Software Engineering, Minneapolis USA
 - Keynote Talk: *Enterprise Architecture Infosys Viewpoint and Capabilities* – 2nd EA Summit, Singapore
- 2006
 - Invited Talk: *Evolution of Infosys - From Service-Based to a Knowledge-Driven Organization* – Dept of ECE, Purdue University USA

PUBLICATIONS

Journal

- [BS25b] Soumyadip Bandyopadhyay and Santonu Sarkar. "Verifying Correctness of PLC Software during System Evolution using Model Containment Approach". In: (Sept. 2025). DOI: 10.48550/ARXIV.2509.05596. arXiv: 2509.05596 [cs.SE].
- [Kum+25] Rupesh Kumar et al. "Reconciling algal growth understanding in photobioreactors through a statistical and facile single parameter (ψ) approach". In: *Environmental Science: Advances* 4.10 (2025), pp. 1650–1662. ISSN: 2754-7000. DOI: 10.1039/d5va00083a.
- [Meh+25] Shlok Mehendale et al. "Radon-Nikodým Derivative: Re-imagining Anomaly Detection from a Measure Theoretic Perspective". In: (Feb. 2025). DOI: 10.48550/ARXIV.2502.18002. arXiv: 2502.18002 [cs.LG].
- [Sin+25] Divyanshu Singh et al. "Altruistic Ride Sharing: A Framework for Fair and Sustainable Urban Mobility via Peer-to-Peer Incentives". In: (Oct. 2025). DOI: 10.48550/ARXIV.2510.13227. arXiv: 2510.13227 [cs.MA].
- [Sah+24] Snehanishu Saha et al. "quantile-Long Short Term Memory: A Robust, Time Series Anomaly Detection Method". In: *IEEE Transactions on Artificial Intelligence* (2024), pp. 1–11. ISSN: 2691-4581. DOI: 10.1109/tai.2024.3353163.

- [Sar+24a] Santonu Sarkar et al. "Can Tree Based Approaches Surpass Deep Learning in Anomaly Detection? A Benchmarking Study". In: (Feb. 2024). DOI: 10.48550/ARXIV.2402.07281. arXiv: 2402.07281 [cs.LG].
- [Ala+23] Gargi Alavani et al. "Program Analysis and Machine Learning based Approach to Predict Power Consumption of CUDA Kernel". In: *ACM Transactions on Modeling and Performance Evaluation of Computing Systems* 8.4 (July 2023), pp. 1–24. ISSN: 2376-3647. DOI: 10.1145/3603533.
- [Sah+23] Snehanshu Saha et al. "Quantile LSTM: A Robust LSTM for Anomaly Detection In Time Series Data". In: (Feb. 2023). DOI: 10.48550/ARXIV.2302.08712. arXiv: 2302.08712 [cs.LG].
- [SSS23] J Sarkar, S Saha, and S Sarkar. "Efficient Anomaly Identification in Temporal and Non-Temporal Industrial Data using Tree Based Approaches". In: *Applied Intelligence* 53.8 (2023), pp. 8562–8595. DOI: 10.1007/s10489-022-03940-3.
- [VBS22] Sreekrishnan Venkateswaran, Adwait Bauskar, and Santonu Sarkar. "Architecture of a time-sensitive provisioning system for cloud-native software". In: *Software: Practice and Experience* 52.5 (2022), pp. 1170–1198. DOI: doi.org/10.1002/spe.3059.
- [AS21] Gargi Alavani and Santonu Sarkar. "Performance modeling of graphics processing unit application using static and dynamic analysis". In: *Concurrency and Computation: Practice and Experience* 34.3 (2021), e6602. DOI: doi.org/10.1002/cpe.6602.
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